

Heron Suspension Service Manual

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1 Suspension System Overview

The Heron suspension uses MacPherson front struts, rear shock absorbers, lower control arms with sway bar links, and pressed wheel hub bearing assemblies. Together these components control ride height, damping, and wheel alignment geometry across the vehicle. The overall layout follows the heavy-duty platform documented in the Ridgeline HD Suspension Service Manual :: Suspension System Overview . Worn dampers or bearings degrade handling and tire wear, so inspect all four corners during any suspension diagnosis.

2 Safety and Service Precautions

Always support the vehicle on rated jack stands and never place any body part under a spring that is compressed by a strut compressor. A loaded front strut STRT-8500 stores significant energy and can cause serious injury if released improperly. Electrical sensors and harnesses routed near the suspension are documented in the Heron Ignition Service Manual , so avoid damaging them during component removal. Torque all fasteners to specification with the suspension loaded to ride height.

3 Front Strut Sub-System

The Heron front strut assembly carries part number STRT-8500 and combines the coil spring, damper, and upper mount into a single serviceable unit. A worn STRT-8500 produces a knocking noise over bumps and allows excessive nose dive under braking. This assembly is functionally equivalent to the unit in the Ridgeline HD Suspension Service Manual :: Front Strut Assembly , though spring rates differ. Always replace front struts in axle pairs to maintain balanced damping.

3.1 Front Strut Replacement

Use procedure PROC-STRUT to remove and install the STRT-8500 assembly after marking the camber bolt position. Compress the spring fully before separating the upper mount, then transfer hardware to the new STRT-8500. The replacement sequence parallels the Ridgeline HD Suspension Service Manual :: Front Strut Replacement Procedure . A four-wheel alignment is required after any front strut replacement.

4 Rear Shock Sub-System

The Heron rear shock absorber carries part number SHK-8520 and provides damping for the rear axle independent of the load-bearing springs. A leaking or gas-depleted SHK-8520 causes a floating ride and poor rebound control. The component is comparable to the damper described in the Ridgeline HD Suspension Service Manual :: Rear Shock Absorber . Inspect the shock body for oil film, which indicates internal seal failure.

4.1 Rear Shock Absorber Replacement

Remove the upper and lower mounting bolts to free the SHK-8520, supporting the axle to prevent strain on the brake lines. Install the new SHK-8520 with fresh hardware and torque the mounts only after the suspension is loaded to ride height. Replace rear shocks as an axle pair to maintain even damping. Road-test the vehicle to confirm the floating ride condition has been resolved.

5 Control Arm and Linkage Sub-System

The Heron lower control arm CARM-8540 locates the front wheel and houses the ball joint and bushings, while the sway bar link SBL-8560 connects the stabilizer bar to the strut. Worn CARM-8540 bushings cause steering wander, and a failed SBL-8560 produces a rattle over rough roads. Both components are detailed alongside

the platform parts in the Ridgeline HD Suspension Service Manual :: Control Arms and Sway Bar Links . Inspect ball joints for play before condemning the control arm.

5.1 Control Arm Replacement

Follow procedure PROC-CONTROL-ARM to replace CARM-8540 after separating the ball joint and removing the pivot bolts. Transfer or replace the bushings as required and torque the pivot bolts with the suspension at ride height. This procedure mirrors the Ridgeline HD Suspension Service Manual :: Control Arm Replacement Procedure . A wheel alignment is mandatory after replacing CARM-8540.

5.2 Sway Bar Link Replacement

Replace the sway bar link SBL-8560 by holding the stud with a wrench while removing the retaining nut to prevent the joint from spinning. Install the new SBL-8560 and torque to specification, ensuring the boot is seated. Refer to the Ridgeline HD Suspension Service Manual :: Control Arm and Sway Bar Sub-System Service for the platform link specifications. Confirm the stabilizer rattle is gone on a short road test.

6 Wheel Hub Bearing Sub-System

The Heron wheel hub bearing assembly WHB-8580 integrates the bearing, hub flange, and ABS reluctor ring into a bolt-on unit. A worn WHB-8580 produces a cyclic droning noise that changes with cornering load. This assembly corresponds to the unit in the Ridgeline HD Suspension Service Manual :: Wheel Hub Bearing Assembly . A damaged reluctor ring on WHB-8580 can also disable ABS function.

6.1 Wheel Hub Bearing Replacement

Use procedure PROC-HUB-BEARING to unbolt and replace the WHB-8580 assembly after removing the brake caliper and rotor. Torque the hub bolts to specification and verify smooth, quiet rotation before reinstalling the brake components. The replacement follows the Ridgeline HD Suspension Service Manual :: Wheel Hub Bearing Sub-System Service . Confirm the wheel speed signal is clean if the WHB-8580 reluctor ring was the source of an ABS fault.